



SESSION 46

Hypothesis Testing Part 2

Student's t-Test · One-Sample, Independent & Paired t-Tests ·
Degrees of Freedom

The Student's t-Test Overview

One-Sample t-Test

Independent Two-Sample t-Test

Paired Sample t-Test

Degrees of Freedom (df)

Assumptions of t-Tests

Normality and Homoscedasticity Checks

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01 • The Student's t-Test Overview

The **t-test** is used to compare means when the population standard deviation (σ) is unknown. It relies on the t-distribution and is widely used for smaller samples.

02 • One-Sample t-Test

Compares the mean of a single sample to a hypothesized population mean (μ_0).

ONE-SAMPLE T-STATISTIC

$$t = (\bar{X} - \mu_0) / (s / \sqrt{n})$$

03 • Independent Two-Sample t-Test

Compares the means of two independent groups (e.g., comparing test scores of two different classrooms).

INDEPENDENT T-STATISTIC (EQUAL VARIANCES)

$$t = (\bar{X}_1 - \bar{X}_2) / [s_p \cdot \sqrt{(1/n_1 + 1/n_2)}] \text{ where } s_p = \sqrt{((n_1-1)s_1^2 + (n_2-1)s_2^2) / (n_1 + n_2 - 2)}$$

04 • Paired Sample t-Test

Compares means from the same group at different times (e.g., weights before and after a diet program). We analyze the differences $D_i = X_{\{1,i\}} - X_{\{2,i\}}$:

PAIRED T-STATISTIC

$$t = \bar{D} / (s_d / \sqrt{n})$$

05 • Degrees of Freedom (df)

Degrees of freedom represent the number of independent values that can vary in an analysis:

- **One-sample / Paired t-test:** $df = n - 1$
- **Independent two-sample t-test:** $df = n_1 + n_2 - 2$

06 • Assumptions of t-Tests

For t-test results to be valid, the following assumptions must hold:

1. **Scale:** Continuous or interval scale of measurement.
2. **Independence:** Random sampling with independent observations.
3. **Normality:** The populations from which samples are drawn should be approximately normal.
4. **Homoscedasticity:** Independent groups should have approximately equal variances (only for two-sample t-tests).

07 · Normality and Homoscedasticity Checks

Use visual checks (Q-Q plots) and formal statistical tests:

- **Normality:** Shapiro-Wilk test (`scipy.stats.shapiro`).
- **Equal Variance:** Levene's test (`scipy.stats.levene`).

08 · Python Code Implementation

```
import numpy as np
from scipy import stats

group_a = [23, 25, 28, 32, 24, 27, 29, 31, 25, 26]
group_b = [30, 32, 35, 31, 29, 33, 34, 32, 31, 30]

# Perform independent two-sample t-test
t_stat, p_val = stats.ttest_ind(group_a, group_b, equal_var=True)
print(f"t-statistic: {t_stat:.4f}, p-value: {p_val:.4f}")
```

09 · Quick Reference Card



t-Test Quick Reference

- **One-sample:** Compare one group mean to a target value.
- **Independent:** Compare means of two distinct groups.
- **Paired:** Compare means of the same group under two conditions.
- **Welch's t-test:** Use `equal_var=False` in Python when group variances differ.